Hemant Sharma

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Education

Ph.D. in Mathematics *2020–2026 (expected)*
Under Supervision of Dr. Nachiketa Mishra
Department of Mathematics, IIITDM Kancheepuram, Chennai
Thesis submitted; CGPA: 8.69/10. Research area: tensors and image processing.

Master of Science in Mathematics *2015–2017*
Department of Mathematics, VNIT Nagpur, Nagpur
CGPA: 6.6/10. Research area: Fixed point theory (fractals).

Bachelor of Science (Hons.) in Mathematics *2012–2015*
PGDAV (Evening) College, University of Delhi, New Delhi
Percentage: 82.23%.

Teaching and Research Experience

Assistant Professor and Head, Department of Mathematics *Sep 2025–Present*
Tungal School of Basic and Applied Sciences
Affiliated to Bagalkot University, Jamkhandi, Karnataka

Teaching Assistant *Jan 2021–May 2025*
Taught Linear Algebra and Numerical Methods to second-year B.Tech. students in Computer Science and Mechanical Engineering in IIITDM Kancheepuram.

Junior Research Fellow *Jan 2019–Feb 2020*
UM–DAE Centre for Excellence in Basic Sciences (CEBS)
Research fellowship funded by the Department of Atomic Energy.

Guest Faculty in Mathematics *Jul 2018–Nov 2018*
PGDAV College, University of Delhi
Taught second-year B.Sc. (Hons.) Mathematics students for one semester.

Publications

1. **Sharma, H.**, and Mishra, N. (2027). A closed-form Ledoit–Wolf shrinkage estimator for T-positive-definite tensor covariance. *Statistics & Probability Letters*, **242**, 111002.
2. **Sharma, H.**, and Mishra, N. (2026). Stein shrinkage for T-product tensor covariance estimation. *Journal of Applied Mathematics and Computing*, **72**, 260.
3. Nayak, S., **Sharma, H.**, and Mishra, N. (2026). Eigenvalue Bounds of Symmetric Positive Definite Tensors. *Communications on Applied Mathematics and Computation*.
4. **Sharma, H.**, and Mishra, N. (2025). Bounding Eigenvalues of Symmetric Positive Definite Tensors: A Comparative Analysis of TDet and TDep Approaches. *Lobachevskii Journal of Mathematics*, **46**(5), 2443–2455.

Awards and Competitive Examinations

- Qualified NET–JRF: All India Rank 32 (December 2017) and 112 (June 2018).
- Recipient of the DAE–NBHM M.A./M.Sc. Scholarship (2015).
- Qualified GATE (2015 and 2018) and JAM (2015).

Academic Training

- **Summer Program in Mathematics (SPIM)**, Harish-Chandra Research Institute, Allahabad, 13 June–2 July 2016.
- **Mathematics Training and Talent Search Programme (MTTS), Level O**, IIT Patna, 9–21 June 2014, conducted by NBHM. Subsequently selected for Level 1.

Leadership and Extracurricular Achievements

- NCC Army Wing: holds 'B' and 'C' certificates; served as Junior Under Officer (2012).
- First place in the mathematics quiz at PGDAV College, University of Delhi, in 2014 and 2015.
- First place in basketball and tug of war at Vihang, VNIT Nagpur's annual sports and cultural meet (2017).
- Participated in the CBSE East Zone III cluster kho-kho competition, under-19 category (2011).